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**State** Finished

**Completed on** Monday, 25 November 2024, 9:23 AM

**Time taken** 57 mins 55 secs

**Grade** Not yet graded

Question **1**

Partially correct

Mark 4.17 out of 5.00

Study the following program (MyClass.java) and select which features does it exemplify. The LoremIpsum part is just a class generating gibberish, it is not much relevant here. Ignore it.

```
1 public class MyClass implements Cloneable {
2     private Object data;
3     private LoremIpsum loremIpsum = new LoremIpsum();
4     private static int toStringCallCount = 0;
5     public MyClass(Object o) {
6         data = o;
7     }
8     protected MyClass clone() throws CloneNotSupportedException {
9         MyClass mc = (MyClass)super.clone();
10        mc.data = new Object();
11        mc.data = data;
12        return mc;
13    }
14    public String toString() {
15        toStringCallCount++;
16        return data.getClass().getSimpleName()+":"+data+": "+
17        loremIpsum.getWords(2,Integer.min(toStringCallCount,40));
18    }
19 }
```

- ☒ Reflection
- ☐ None
- ☐ Functional
- ☐ Procedural
- ☐ Deep copy
- ☒ Shallow copy



Your answer is partially correct.

You have correctly selected 2.

The correct answers are: Reflection

,

Deep copy,

Shallow copy

## Question 2

Incorrect

Mark 0.00 out of 5.00

Study the following main program that instantiates the MyClass seen in the first question in an array. (It is again reproduced below for ref.) What makes it throw CloneNotSupportedException?

```

1 import java.util.*;
2
3 public class Main {
4     public static void main(String argv[]) throws CloneNotS
5         ArrayList<MyClass> l1 = new ArrayList<MyClass>();
6         ArrayList<MyClass> l3 = new ArrayList<MyClass>();
7         ArrayList<MyClass> l2 = l1;
8         long seed = 1234567890;
9         int size = 10;
10        if(argv.length>0) size = Integer.valueOf(argv[0]);
11        Random rand = new Random(seed);
12        for(int i = 0; i < size; i++) l1.add(new MyClass(rand
13        for(int i = 0; i < size; i++) l3.add(new MyClass(l1.g
14        for(int i = 0; i < size; i++) {
15            System.out.print(l1.get(i).getClass().getName()+": "
16            System.out.print(l2.get(i).getClass().getName()+": "
17            System.out.print(l3.get(i).getClass().getName()+": "
18        }
19        System.out.println();
20    }
21 }

1 public class MyClass implements Cloneable {
2     private Object data;
3     private LoremIpsum loremIpsum = new LoremIpsum();
4     private static int toStringCallCount = 0;
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9         MyClass mc = (MyClass)super.clone();
10        mc.data = new Object();
11        mc.data = data;
12        return mc;
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14    public String toString() {
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16        return data.getClass().getSimpleName()+":"+data+": "+
17        loremIpsum.getWords(2,Integer.min(toStringCallCount,40
18    }
19 }

```

Answer: There is nothing in MyClass that would throw CloneNotSupportedException. The "thi 

## Question 3

Incorrect

Mark 0.00 out of 5.00

Study the following main and myclass programs and give a short formula for the number of occurrences of MyClass in the output when we run (after due compilation) in terms of  $n$

java Main  $n$

where  $n$  is a positive integer. The answer format is expected to be something like " $kn$ " where  $k$  is a fixed constant, or " $kn^2$ ", " $k_1n + k_2n^2$ " etc.

```

1 import java.util.*;
2
3 public class Main {
4     public static void main(String argv[]) throws CloneNotS
5         ArrayList<MyClass> l1 = new ArrayList<MyClass>();
6         ArrayList<MyClass> l3 = new ArrayList<MyClass>();
7         ArrayList<MyClass> l2 = l1;
8         long seed = 1234567890;
9         int size = 10;
10        if(argv.length>0) size = Integer.valueOf(argv[0]);
11        Random rand = new Random(seed);
12        for(int i = 0; i < size; i++) l1.add(new MyClass(rand
13        for(int i = 0; i < size; i++) l3.add(new MyClass(l1.g
14        for(int i = 0; i < size; i++) {
15            System.out.print(l1.get(i).getClass().getName()+": "
16            System.out.print(l2.get(i).getClass().getName()+": "
17            System.out.print(l3.get(i).getClass().getName()+": "
18        }
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1 public class MyClass implements Cloneable {
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8     protected MyClass clone() throws CloneNotSupportedException
9         MyClass mc = (MyClass)super.clone();
10        mc.data = new Object();
11        mc.data = data;
12        return mc;
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14    public String toString() {
15        toStringCallCount++;
16        return data.getClass().getSimpleName()+":"+data+": "+
17        loremIpsum.getWords(2,Integer.min(toStringCallCount,40
18    }
19 }

```

Answer: 3n



The correct answer is: 4 n

## Question 4

Partially  
correctMark 3.75 out  
of 5.00

You are to create a search tree (AVL, Red-Black, 2-4-Trees, etc.) class hierarchy. For keeping the trees balanced, we need to update global and local properties (like heights and balancing, of the whole and subtrees) in each node and subtree repeatedly. Also balancing requires restructuring the tree, sometimes even changing the root. The application may involve data in each node being a stream, a dynamic object that grows and shrinks without being in the main memory completely. Moreover, each node's key may change dynamically and asynchronously in response to the data stream flow, changing its position in the order, in turn requiring re-balancing of the tree.

Will you create your own binary tree class

```
class BinTree { private BinTree left = null, right = null; int height = 0; Key key; ...}
```

as the base class? Or it is easier to use container classes like maps treemaps etc.? Compare and select the true considerations given below.

- ☐ Both implementations have the same behaviour for global properties and restructuring. There is nothing to choose from.
- ☒ The first (BinTree) implementation requires deep copies. That makes restructuring cumbersome if it is not to be hazardous. ✓
- ☒ The second (map containers) is less cumbersome but synchronization is still complicated. ✓
- ☐ Handling dynamic keys is the same for both the implementations.
- ☐ None of the options is relevant or true.

Your answer is partially correct.

You have correctly selected 2.

The correct answers are:

The first (BinTree) implementation requires deep copies. That makes restructuring cumbersome if it is not to be hazardous.,

The second (map containers) is less cumbersome but synchronization is still complicated.,

Handling dynamic keys is the same for both the implementations.

## Question 5

Complete

Marked out of  
5.00

Study the following CLOS program and answer the following question.

```
(defclass figure ()
  (
    (height :initform 0.0 :initarg :height)
    (width :initform 0.0 :initarg :width)
  )
)
(defclass hue-mixture ()
  (
    (cyan :initform 0 :initarg :cyan)
    (magenta :initform 0 :initarg :magenta)
    (yellow :initform 0 :initarg :yellow)
  )
)
(defclass paint-figure (hue-mixture figure)
  (
    (clearp :initform (y-or-n-p "But is it transparent?")
             :initarg :clearp :accessor clearp)
  )
)
(defgeneric render (x))
(defmethod render ((x figure)) ;Method #1
  (vertical-stroke
   (slot-value x 'height)
   (slot-value x 'width)
  )
)
(defmethod render :before ((x hue-mixture)) ;Method #2
  (set-brush-hue
   (slot-value x 'cyan)
   (slot-value x 'magenta)
   (slot-value x 'yellow)
  )
)
(defmethod render ((x paint-figure)) ;Method #3
  (unless (clearp x) (call-next-method))
)
(setq another-figure (make-instance 'figure :width 10.0 :height 10.0))
(setq some-figure
  (make-instance 'paint-figure
    :width 38 :height 84 :cyan 60
    :yellow 55 :clearp nil
  )
)
(defun vertical-stroke (height width)
  (print (list height width))
)
)
```

Now, after loading (and compiling) the above program only, at the top level in the interpreter shell, the following evaluations are sought. Explain what do you expect to happen and why.

```
(render another-figure)
```

```
(render some-figure)
```

 [2022A7PS0244G-POPLQ2.pdf](#)

For **another-figure** it prints the rectangle size. For **some-figure** there is an "MISSING FDEFINITION SET-BRUSH-HUE" run-time error. If it is supplied, it again prints the size.

